We went over Doppler shift pretty quickly so I want to begin today with a few examples.

$\Rightarrow$ Top Hat Q1 and Q2

More challenging example:

A dolphin is moving at speed v_d and emits an echolocation sound at frequency f. The sound reflects of a stationary object and returns to the dolphin. At what frequency will the sound be heard?



There are two parts to this problem: original sound f being received by the object, then the object acting as the source when the sound bounces off.

Part 1: $S \rightarrow D$ with $v_S = v_d$, $v_D = 0$

$f' = \left(\frac{v}{v - v_d}\right) f$ is the frequency that hits the object.

Part 2: The sound f' bounces off with no shift in frequency, so now the object becomes the source at freq. f':

$D \rightarrow S$ with $v_S = 0$, $v_D = v_d$

$$f'' = \left(\frac{v + v_d}{v}\right) f' = \left(\frac{v + v_d}{v}\right)\left(\frac{v}{v - v_d}\right) f$$

$$\Rightarrow \boxed{f'' = \left(\frac{v + v_d}{v - v_d}\right) f}$$

Now we will begin looking at light, which is an electromagnetic wave.

$\Rightarrow$ slides for EM waves

EM waves behave the same as every other wave we have talked about except for one important difference: there is no preferred rest frame. This means the Doppler shift formula is different: $f' = \left(\frac{v \pm v_D}{v \pm v_S}\right) f$ does <u>not</u> apply. A treatment of this for light requires relativity and is <u>not</u> part of PHYS 369.

Light sources typically produce a range of frequencies with waves that carry a mixture of phase constants that are random. This is why we do not notice interference effects between two lamps that are turned on. To get interference effects we need to use <u>coherent</u> sources:

- monochromatic (single frequency)
- constant phase relation between them (each source has a set phase, not random distribution of them).

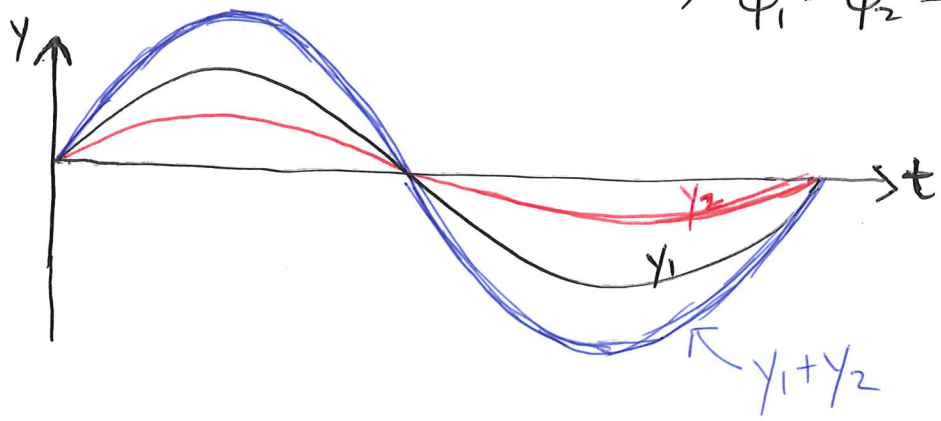
An example of such a source is laser light, which is so coherent that the light you see in the dot has some structure to it, which is an interference pattern on your retina.

We will assume that we have coherent sources for what follows. If two coherent waves add together:

$$y_1 = A_1 \cos(kx - \omega t + \phi_1)$$
$$y_2 = A_2 \cos(kx - \omega t + \phi_2)$$

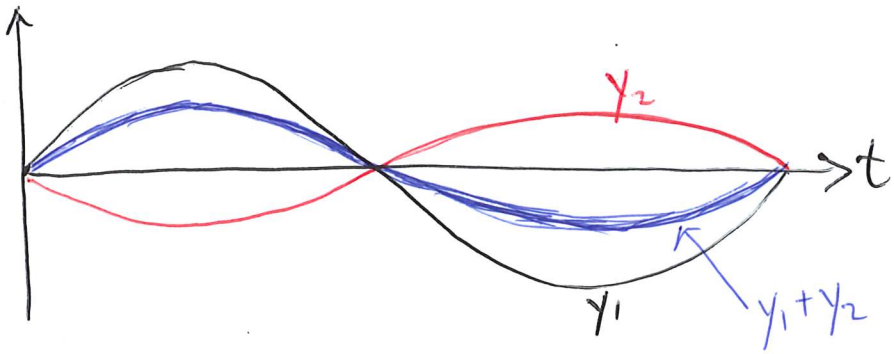
They are <u>in phase</u> when $kx - \omega t + \phi_1 = kx - \omega t + \phi_2 + 2\pi m$

$$\Rightarrow \phi_1 - \phi_2 = 2\pi m$$



The waves <u>constructively interfere</u>; amplitude is $A_1 + A_2$

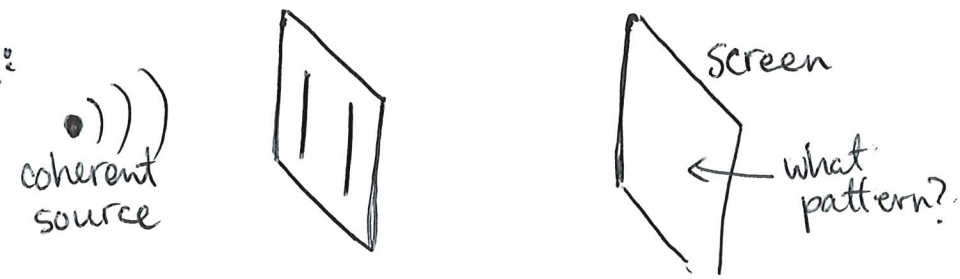
They are <u>out of phase</u> when $\phi_1 - \phi_2 = \pi + 2\pi m$



The waves <u>destructively interfere</u>: amplitude is $A_1 - A_2$

This holds true even when the waves are not travelling in 1D together. Wherever the waves meet, their phase difference determines what type of interference is produced.

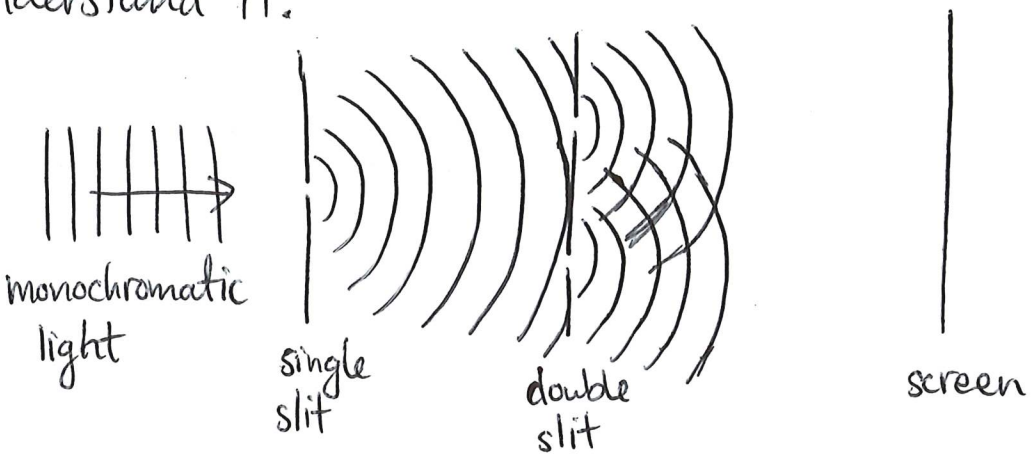
<u>Two slit interference:</u>



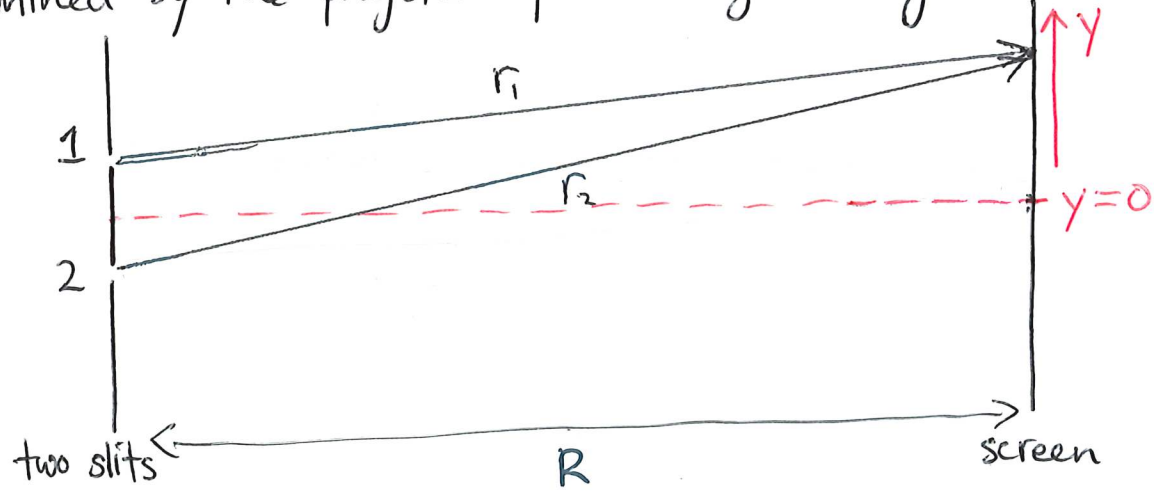
⇒ video clip

5 minute break

Let's look at this setup and figure out how to understand it.



monochromatic light is sent through a single slit, which then acts as a point source as the light diffracts through. This then ensures the light hitting each of the slits on the next partition emerge in phase with each other. Each of the two slits then act as a point source and both sources are in phase with each other. When light from the two sources hits the screen a distance R away, the two waves have a phase determined by the physical path length they have travelled:



The phase of wave 1 is $kr_1 - \omega t + \phi$
The phase of wave 2 is $kr_2 - \omega t + \phi$
} ϕ is the same for both by design

$\Rightarrow$ constructive interference when:

$$kr_1 - \cancel{\omega t} + \cancel{\phi} = kr_2 - \cancel{\omega t} + \cancel{\phi} + 2\pi m$$

$$r_1 - r_2 = \frac{2\pi m}{k} \quad \text{but} \quad k = \frac{2\pi}{\lambda}$$

$$\Rightarrow \boxed{r_1 - r_2 = m\lambda}$$ condition for constructive interference

$m = 0, \pm 1, \pm 2, \ldots$

$\Rightarrow$ destructive interference when:

$$kr_1 - \cancel{\omega t} + \cancel{\phi} = kr_2 - \cancel{\omega t} + \cancel{\phi} + \pi + 2\pi m$$

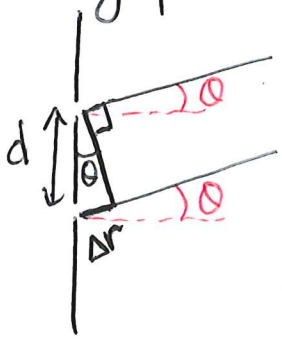
$$r_1 - r_2 = \frac{\pi + 2\pi m}{k} \quad \text{but} \quad k = \frac{2\pi}{\lambda}$$

$$\boxed{r_1 - r_2 = \left(\frac{1}{2} + m\right)\lambda}$$ condition for destructive interference.

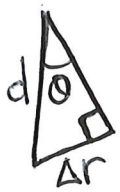
$m = 0, \pm 1, \pm 2, \ldots$

This leads to a series of bright (constructive) and dark (destructive) interference fringes on the screen.

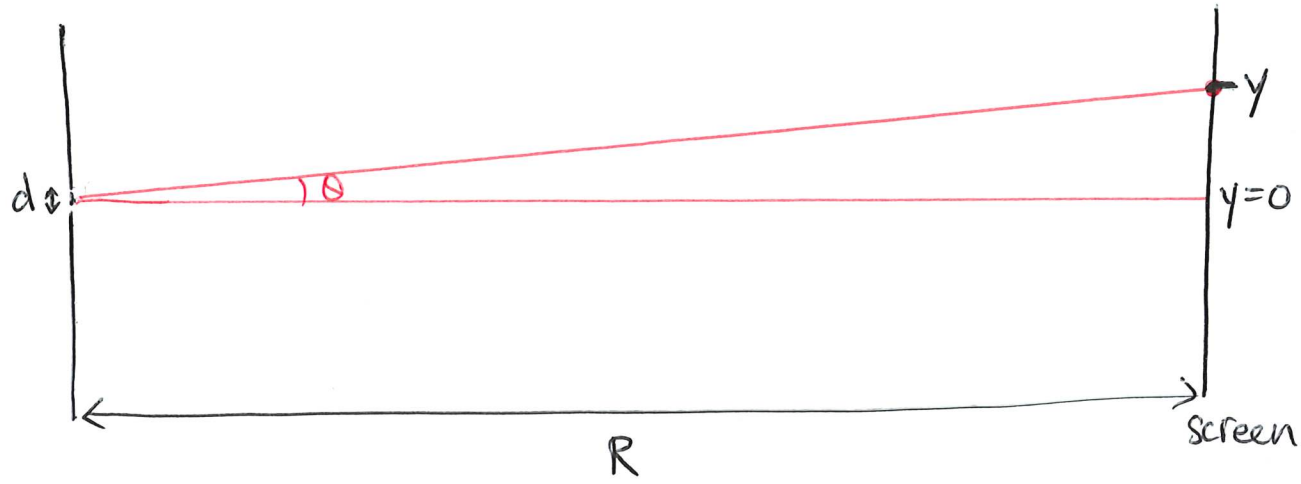
The slits are a distance d apart and the screen is located a distance $R \gg d$ from the slits, so the two light waves are approximately parallel, which simplifies the calculation:



We get Δr from the right triangle



$$\Delta r = d \sin\theta$$



The location on the screen, y, is related to angle, θ, and the distance to the screen, R:

$$\tan\theta = \frac{y}{R} \qquad \text{for small angles } \tan\theta \approx \sin\theta$$

Small angle approximation: $\sin\theta = \frac{y}{R}$ so $\Delta r = y\frac{d}{R}$

Bright fringes appear at $\Delta r = m\lambda \Rightarrow \boxed{y_m = m\left(\frac{R\lambda}{d}\right)}$ $m = 0, \pm 1, \pm 2, \ldots$

Dark bands appear at $\Delta r = \left(\frac{1}{2}+m\right)\lambda \Rightarrow \boxed{y_m = \left(\frac{1}{2}+m\right)\left(\frac{R\lambda}{d}\right)}$ $m = 0, \pm 1, \pm 2, \ldots$

Note: the intensity diminishes as m increases, so the small angle approximation is usually valid.

Example: light from a HeNe laser ($\lambda = 632.8$ nm) is incident on 2 slits separated by 0.125 mm. What is the distance between the $m=-2$ and $m=2$ bright fringes on a screen 3.70m away?

This is constructive interference, so use $y_m = m\frac{R\lambda}{d}$

For $m=-2$: $y_{-2} = (-2)\left(\frac{(3.7\text{m})(632.8\times10^{-9}\text{m})}{0.125\times10^{-3}\text{m}}\right) = -37.5\text{mm}$

$m=+2$ will be at $y_2 = +37.5\text{ mm}$

Therefore these fringes are separated by

$$\boxed{\Delta y = 75.0\text{ mm}}$$